d
```
SELECT Name, Class
  FROM TEACHER
  ORDER BY Class;
```

10 Boolean logic

1 **a** OR gate

b NAND gate

c XOR gate

2

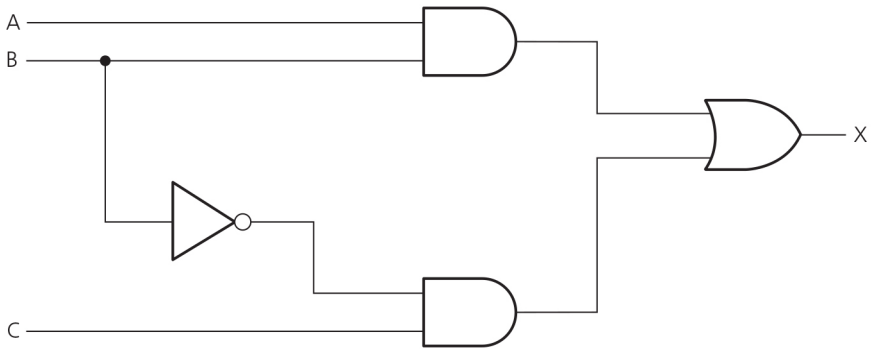
A	B	C	Working area	X
0	0	0		**0**
0	0	1		**0**
0	1	0		**0**
0	1	1		**1**
1	0	0		**1**
1	0	1		**1**
1	1	0		**0**
1	1	1		**1**

3 **a** `(A AND B) OR (NOT B AND C)`

or

`(A.B) + (`$\overline{B}$`.C)`

b



4 **a**

A	B	Working area	X
0	0		**0**
0	1		**1**
1	0		**1**
1	1		**1**

b OR gate

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5 a

A	B	C	Working area	X
0	0	0		**0**
0	0	1		**1**
0	1	0		**0**
0	1	1		**1**
1	0	0		**0**
1	0	1		**1**
1	1	0		**0**
1	1	1		**1**

b As the output X exactly matches input C, the entire logic circuit could be removed and the output connected directly to the input C.

6 a

A	B	C	Working area	X
0	0	0		**1**
0	0	1		**1**
0	1	0		**0**
0	1	1		**0**
1	0	0		**1**
1	0	1		**1**
1	1	0		**0**
1	1	1		**1**

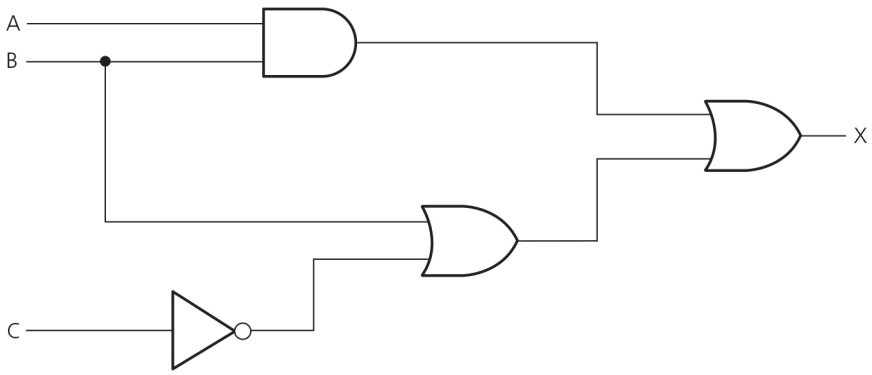
b AND gate

7 a `(A AND NOT B) AND (B OR C)`

or

$(A.\overline{B}).(B + C)$

b



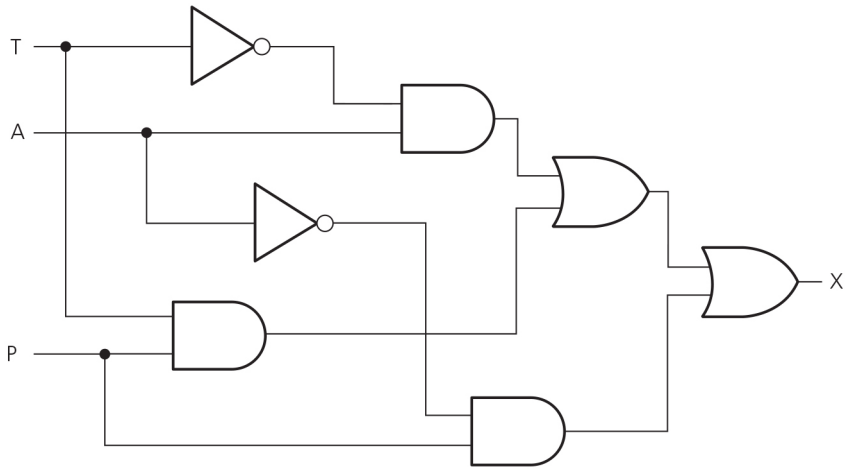
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8 a ((NOT T AND A) OR (T AND P)) OR (NOT A AND P)

or

$(\overline{T}.A + T.P) + (\overline{A}.P)$

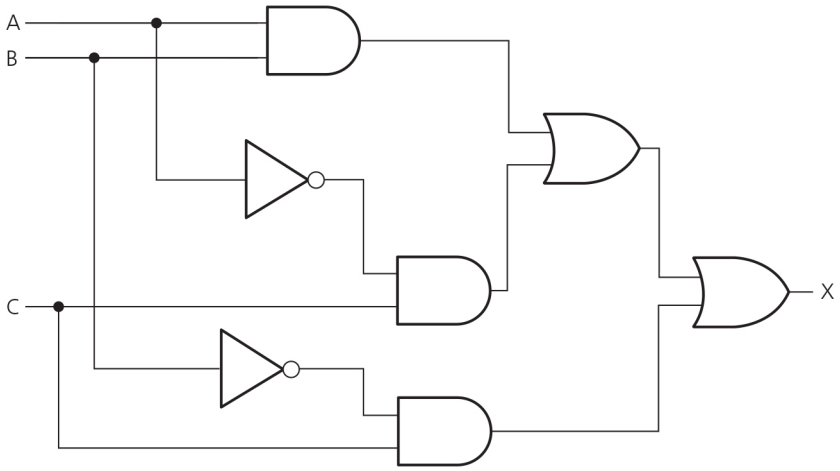
b



c

T	A	P	Working area	X
0	0	0		**0**
0	0	1		**1**
0	1	0		**1**
0	1	1		**1**
1	0	0		**0**
1	0	1		**1**
1	1	0		**0**
1	1	1		**1**

9 a



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b

A	B	C	Working area	X
0	0	0		**0**
0	0	1		**1**
0	1	0		**0**
0	1	1		**1**
1	0	0		**0**
1	0	1		**1**
1	1	0		**1**
1	1	1		**1**

c

X	Y	Z	Working area	Q
0	0	0		**0**
0	0	1		**0**
0	1	0		**0**
0	1	1		**1**
1	0	0		**0**
1	0	1		**0**
1	1	0		**1**
1	1	1		**1**

10 a `((A AND B) AND NOT A) OR (B OR NOT C)`

or

$(A.B).\overline{A} + (B + \overline{C})$

A	B	C	Working area	X
0	0	0		**1**
0	0	1		**0**
0	1	0		**1**
0	1	1		**1**
1	0	0		**1**
1	0	1		**0**
1	1	0		**1**
1	1	1		**1**

11 a `(A AND B) OR (NOT A AND NOT B)`

or

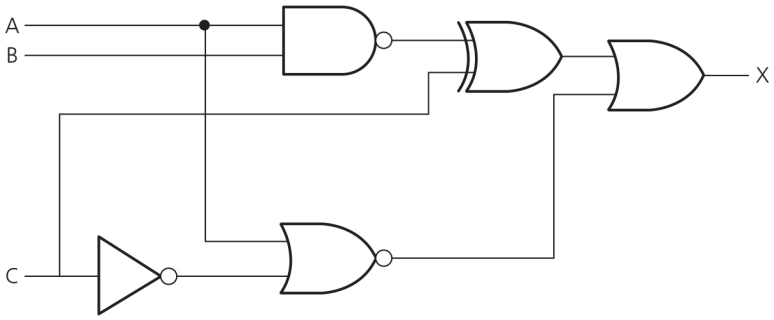
$(A.B) + (\overline{A}.\overline{B})$

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b



12 a



b

A	B	C	Working area	X
0	0	0		**1**
0	0	1		**1**
0	1	0		**1**
0	1	1		**1**
1	0	0		**1**
1	0	1		**0**
1	1	0		**0**
1	1	1		**1**

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